

Festival

Nayra ari gukina umukino mu kirori aho ashobora gutsindira igihembo cyo kujya muri Laguna Colorada.

Yatagiye umukino afite A tokens (urupapuro rutuma ahaha ikintu runaka), ashobora gukoresha agura coupons (urupapuro rutuma agura ku giciro gito). Hari N coupons, ziriho ikimenyetso guhera kuri 0 kugera kuri $N - 1$. Coupon i igura $P[i]$ tokens ($0 \leq i < N$). Ku bwizo mpamvu, Nayra agomba kwishura $P[i]$ tokens kugira ngo ayigire (kandi agomba kuba afite nibura $P[i]$ tokens mbere yo guhaha). Ashobora kugura coupon imwe inshuro zitarenze imwe.

Ikindi kandi, buri coupon i ($0 \leq i < N$) igiye ifite ikimenyetso cy' **ubwoko**, cyerekanwa n' umubare $T[i]$, ni umubare **uri hagati ya 1 na 4, inclusive**. Nyuma yuko Nayra agura coupon i , tokens zose yarasigaranye zikuba inshuro $T[i]$. Muri make, niba afite X mu gihe runaka mu mukino, hanyuma akagura coupon i (isaba ko $X \geq P[i]$), azasigarana $(X - P[i]) \cdot T[i]$ tokens nyuma yo guhaha.

Akazi kawe ni ukwerekana coupons Nayra agomba kugura detse nuko agomba kubikurikinya, kugira ngo aze kuragiza afite **coupons** nyinshi zishoboka. Mu gihe hari uburyo bwinshi ashobora kubikoramo, wakerekana bumwe muri bwo.

Implementation Details

Ugomba gukora iyi function.

```
std::vector<int> max_coupons(int A, std::vector<int> P,  
                             std::vector<int> T)
```

- A : umubare wa tokens Nayra yatagiranye.
- P : array ifite length N yerekana ibiciro bya coupons.
- T : array ifite length N yerekana ubwoko bwa coupons.
- Iyi function ihamagarwa inshuro imwe kuri buri test case.

Iyo function igomba ku returninga array R , yerekana ibyo Nayra yaguze nkuko bisobanuwe hano:

- Length ya R igomba kuba ingana n' umubare munini ushoboka wa coupons ashobora kugura.
- Elements ziri muri array ni indices za coupons agomba kugura, uko zigenda zikurikirana. Bishatse kuvuga, agura coupon $R[0]$ bwambere, coupon $R[1]$ bwakabiri, gutyo gutyo.

- Elements zose za R zigomba kuba zitandukanye.

Mu gihe nta coupons ashobora kugura, array R igomba kuba irimo ubusa.

Constraints

- $1 \leq N \leq 200\,000$
- $1 \leq A \leq 10^9$
- $1 \leq P[i] \leq 10^9$ kuri buri i aho $0 \leq i < N$.
- $1 \leq T[i] \leq 4$ kuri buri i aho $0 \leq i < N$.

Subtasks

Subtask	Score	Ibisabwa
1	5	$T[i] = 1$ kuri buri i aho $0 \leq i < N$.
2	7	$N \leq 3000$; $T[i] \leq 2$ kuri buri i aho $0 \leq i < N$.
3	12	$T[i] \leq 2$ kuri buri i aho $0 \leq i < N$.
4	15	$N \leq 70$
5	27	Nayra ashobora kugura N coupons zose (abitondekanyije mu buryo runuka).
6	16	$(A - P[i]) \cdot T[i] < A$ kuri buri i aho $0 \leq i < N$.
7	18	Ntabindi bisabwa.

Examples

Example 1

Nk' urugero.

```
max_coupons(13, [4, 500, 8, 14], [1, 3, 3, 4])
```

Nayra mbere yarafite tokens $A = 13$. Ashobora kugura 3 coupons mu buryo bwerekanwe haruguru:

Coupon zaguzwe	Igiciro cya Coupon	ubwoko bwa coupon	Umubare wa tokens nyuma yo guhaha
2	8	3	$(13 - 8) \cdot 3 = 15$
3	14	4	$(15 - 14) \cdot 4 = 4$
0	4	1	$(4 - 4) \cdot 1 = 0$

Muri uru rugero, ntago bishoboka ko Nayra yagura coupons zirenze 3, kandi uburyo yabitondekanyije ni bwo bwonyine bwatuma agura 3. Bityo rero, function ingomba kureturninga $[2, 3, 0]$.

Example 2

Tekereza kuri uru rugero.

```
max_coupons(9, [6, 5], [2, 3])
```

Muri uru rugero, Nayra ashobora kugura coupons zose abitondekanyije mu buryo ashatse. Bityo rero, function ishobora kureturninga $[0, 1]$ cyangwa $[1, 0]$.

Example 3

Fata uru rugero.

```
max_coupons(1, [2, 5, 7], [4, 3, 1])
```

Muri uru rugero, Nayra afite token imwe kandi nta coupon nimwe ishobora kugura. Bityo rero, iyi function ishobora kureturninga array $[]$ irimo ubusa.

Sample Grader

Input format:

```
N A
P[0] T[0]
P[1] T[1]
...
P[N-1] T[N-1]
```

Output format:

```
S  
R[0]  R[1]  ...  R[S-1]
```

Hano S ni length ya array R yagaruwe na function `max_coupons`.